

Multiple Choice (10 marks)

1. A _____ is a method in which a computer security mechanism is bypassed untraceable for accessing the computer or its information.
 - (a) front-door
 - (b) backdoor
 - (c) clickjacking
 - (d) key-logging
2. Which of the following is NOT a reasonable justification for choosing to busy-wait on an asynchronous event?
 - (a) The wait is expected to be short.
 - (b) A busy-wait loop is easier to code than an interrupt handler.
 - (c) There is no other work for the processor to do.
 - (d) The program executes on a time-sharing system.
3. Which of the following regular expressions is equivalent to (describes the same set of strings as) $(a^* + b)^*(c + d)$?
 - (a) $a^*(c + d) + b(c + d)$
 - (b) $a^*(c + d)^* + b(c + d)^*$
 - (c) $a^*(c + d) + b^*(c + d)$
 - (d) $(a + b)^*c + (a + b)^*d$
4. What was the first buffer overflow attack?
 - (a) Love Bug
 - (b) SQL Slammer
 - (c) Morris Worm
 - (d) Code Red
5. The Message confidentiality or privacy means that the sender and the receiver expect
 - (a) Integrity
 - (b) Confidentiality
 - (c) Authentication
 - (d) Nonrepudiation

True / False (10 marks)

Answer True or False

6. True or False: Interrupts are generated primarily when data needs to be transferred over a network, rather than for I/O completion.
7. True or False: Maltego is primarily designed for wireless traffic analysis and network sniffing.
8. True or False: Identifier length is best specified using a context-free grammar in programming languages.
9. True or False: WPA is the central node of 802.11 wireless operations, managing network security and access control.
10. True or False: A heap area is typically not included in a subroutine's activation record frame in stack-based programming languages.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to convert a given tuple of positive integers into an integer.
12. Write a function to check if the given array represents min heap or not.

End of Paper